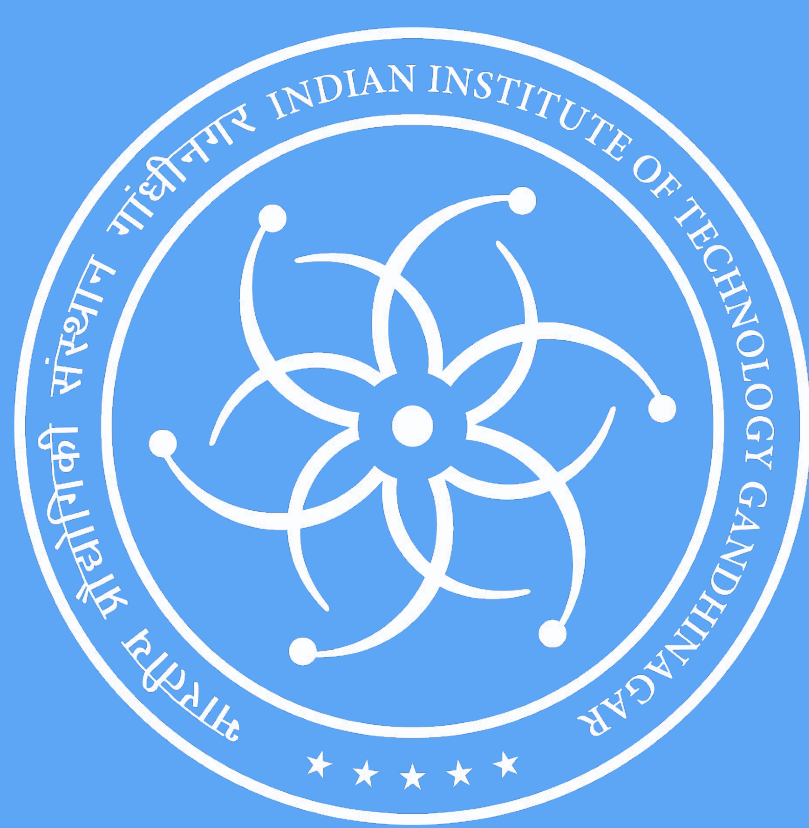




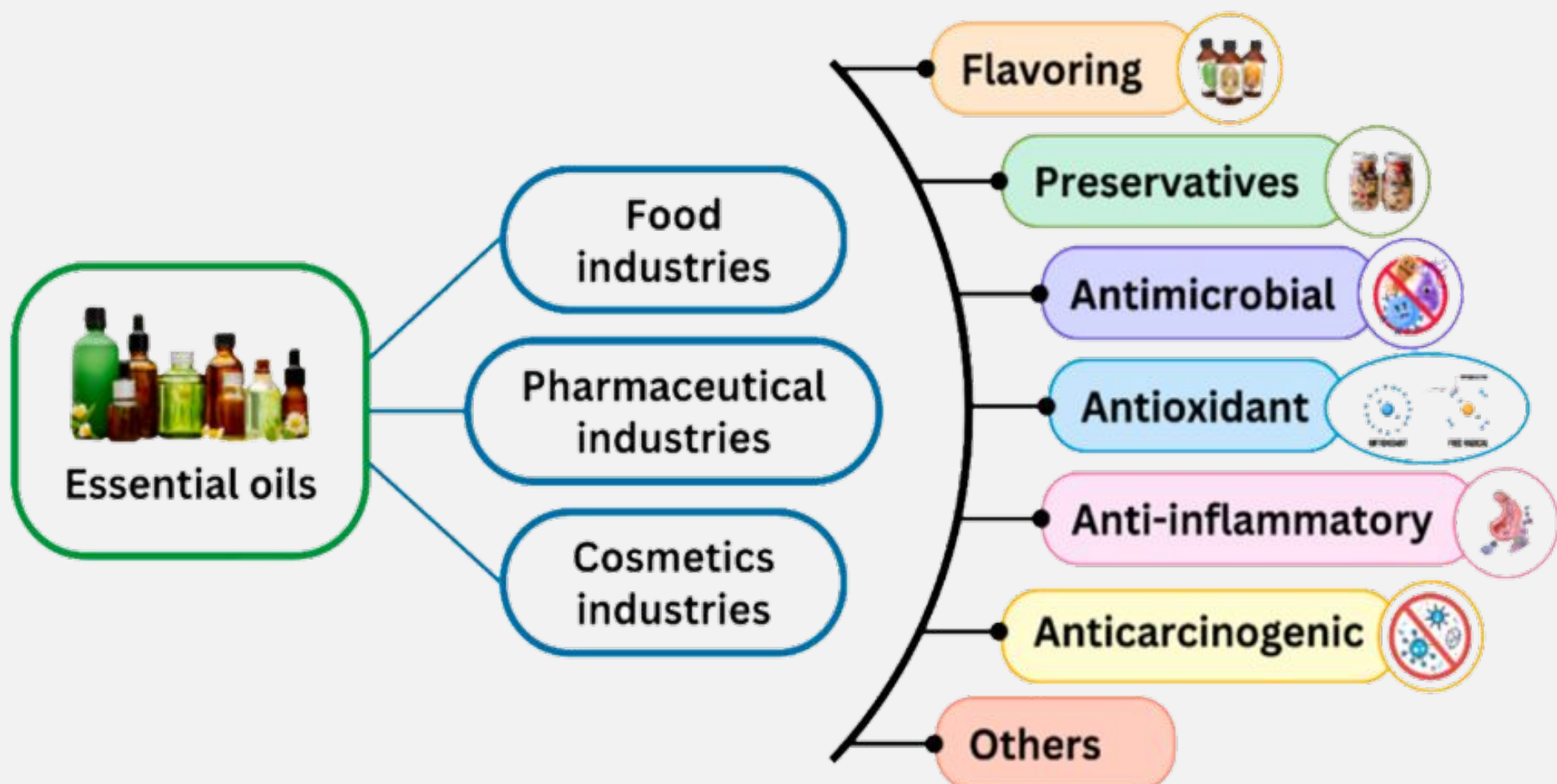
Essential Oil Extraction

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Undergraduate
Research
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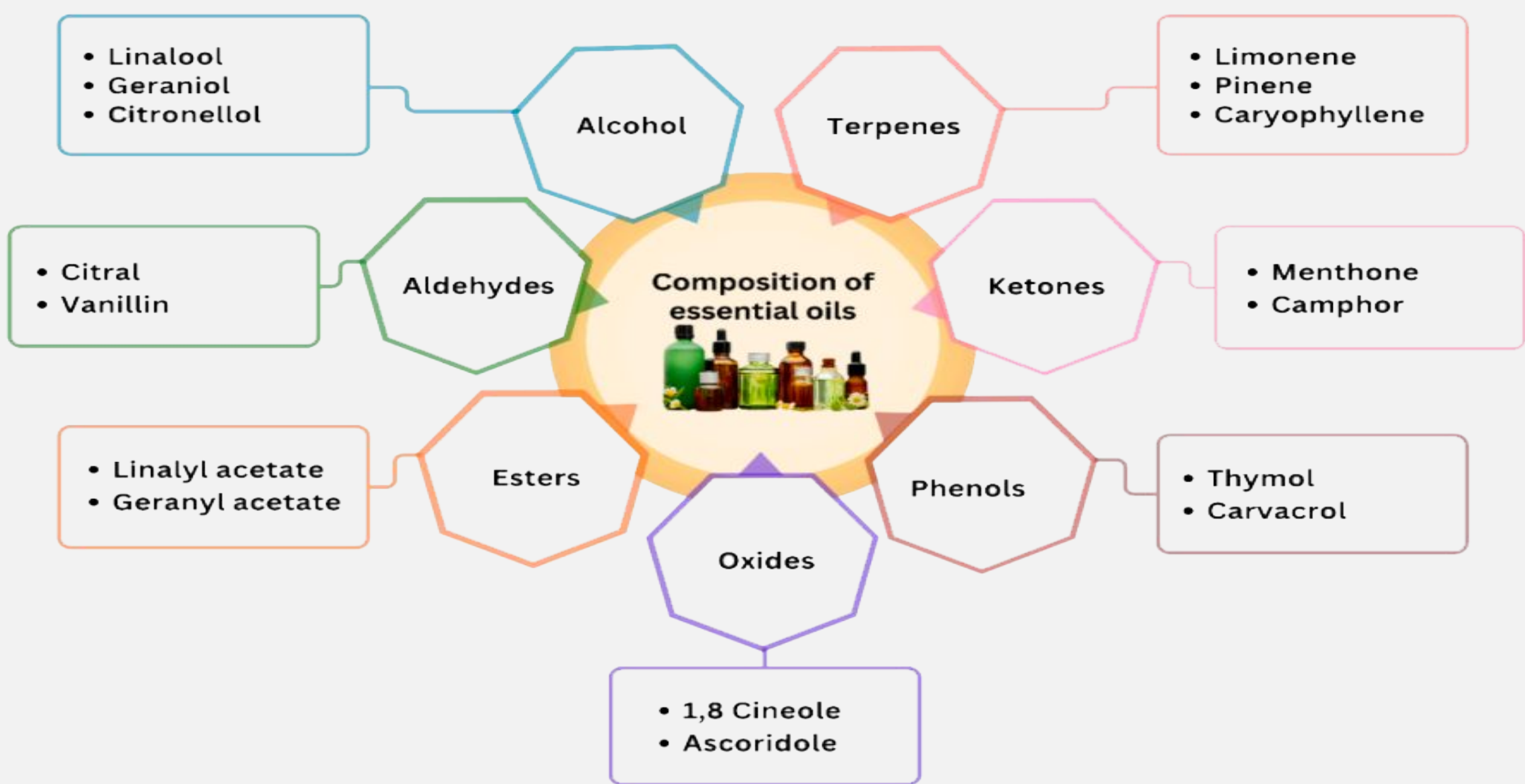
Introduction

- The experiment aims to extract essential oil from fresh lemongrass using steam distillation.
- Steam is passed through chopped lemongrass to break the oil bearing cells and release volatile components.
- The mixture of steam and essential oil is directed to a condenser for cooling.
- On condensation, the mixture separates into two layers, allowing the oil to be collected from the water phase.
- Extraction time, steam flow, feed preparation, and overall oil yield were studied.
- The experiment helps understand the efficiency and practical operation of a laboratory scale steam distillation unit.

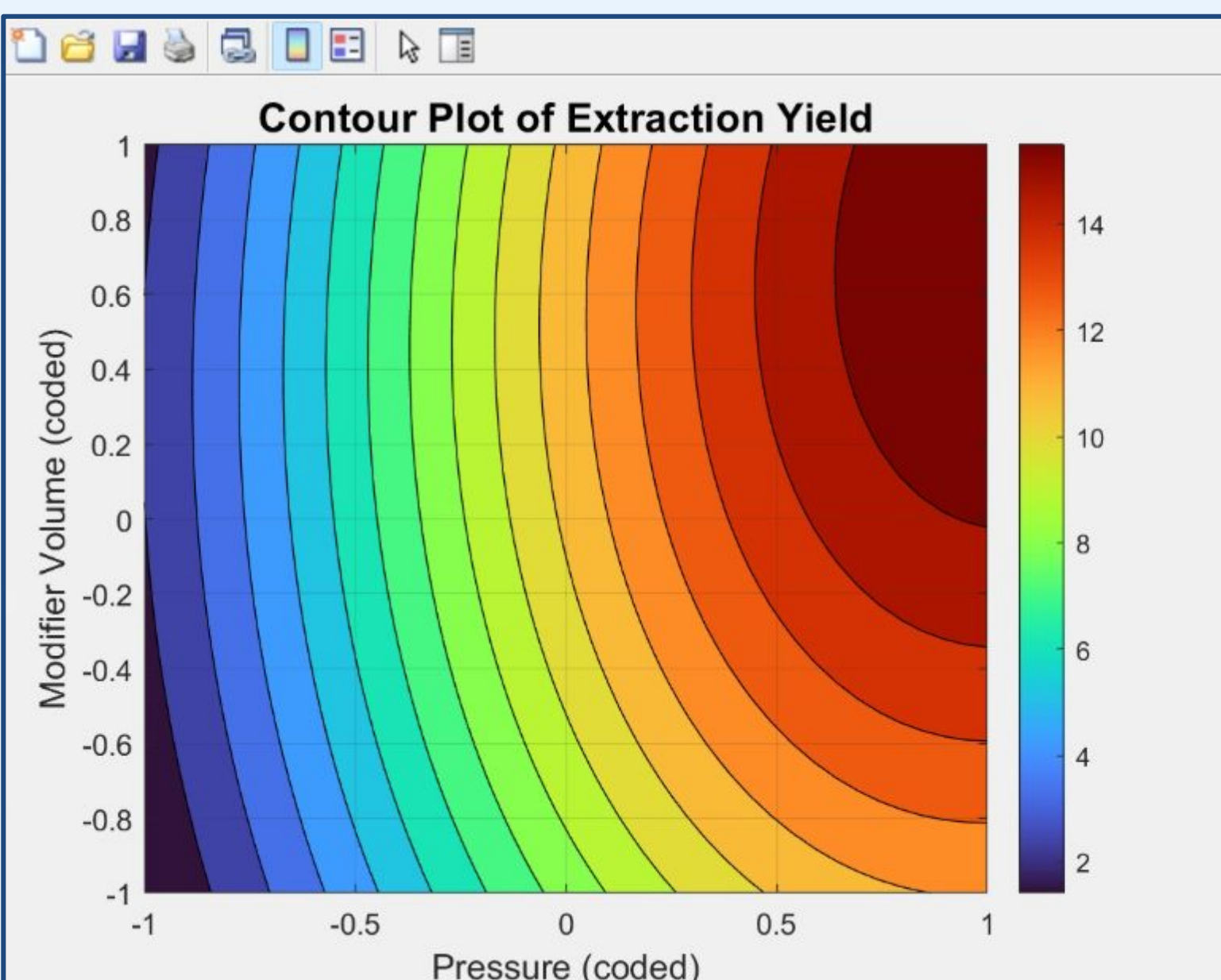
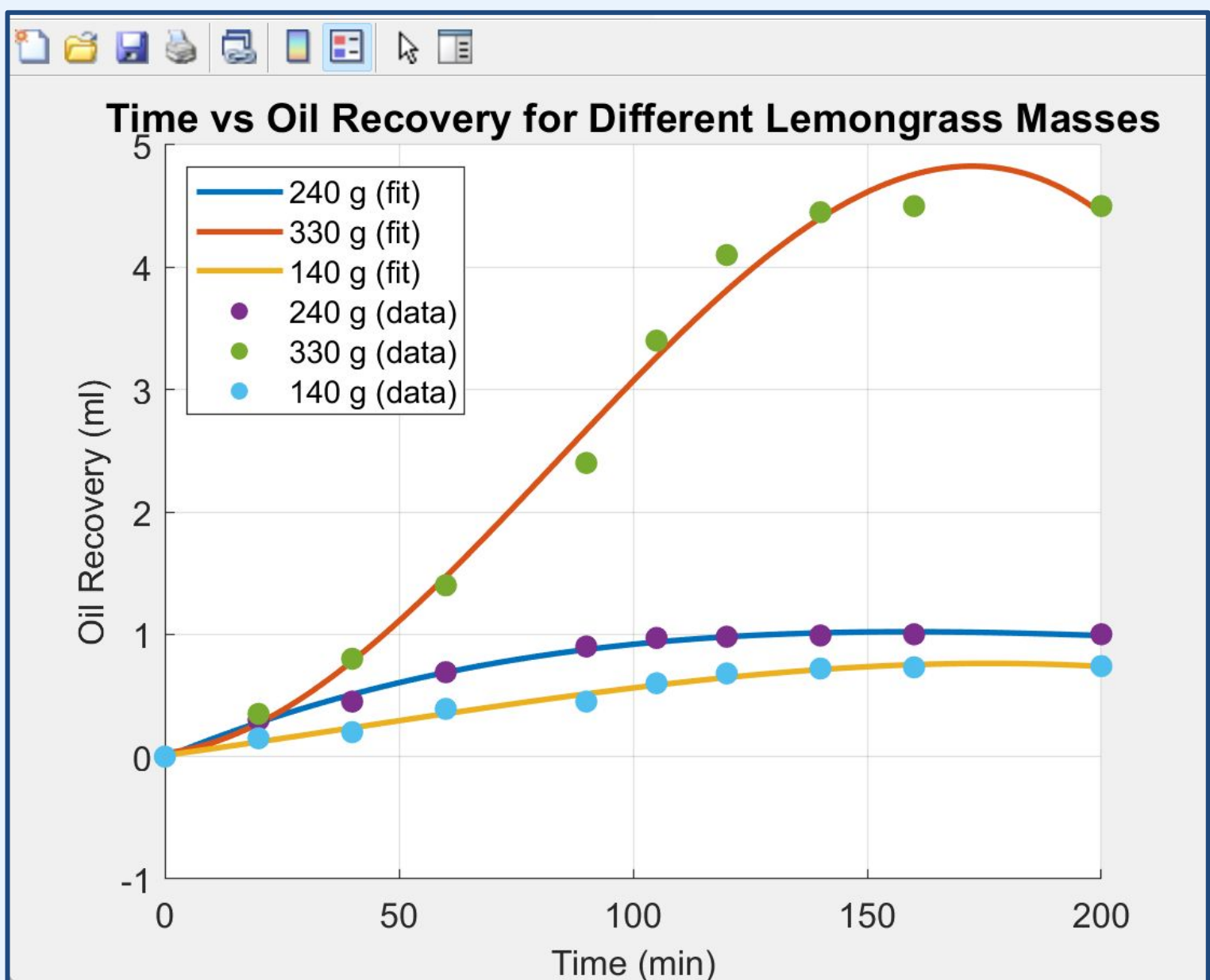
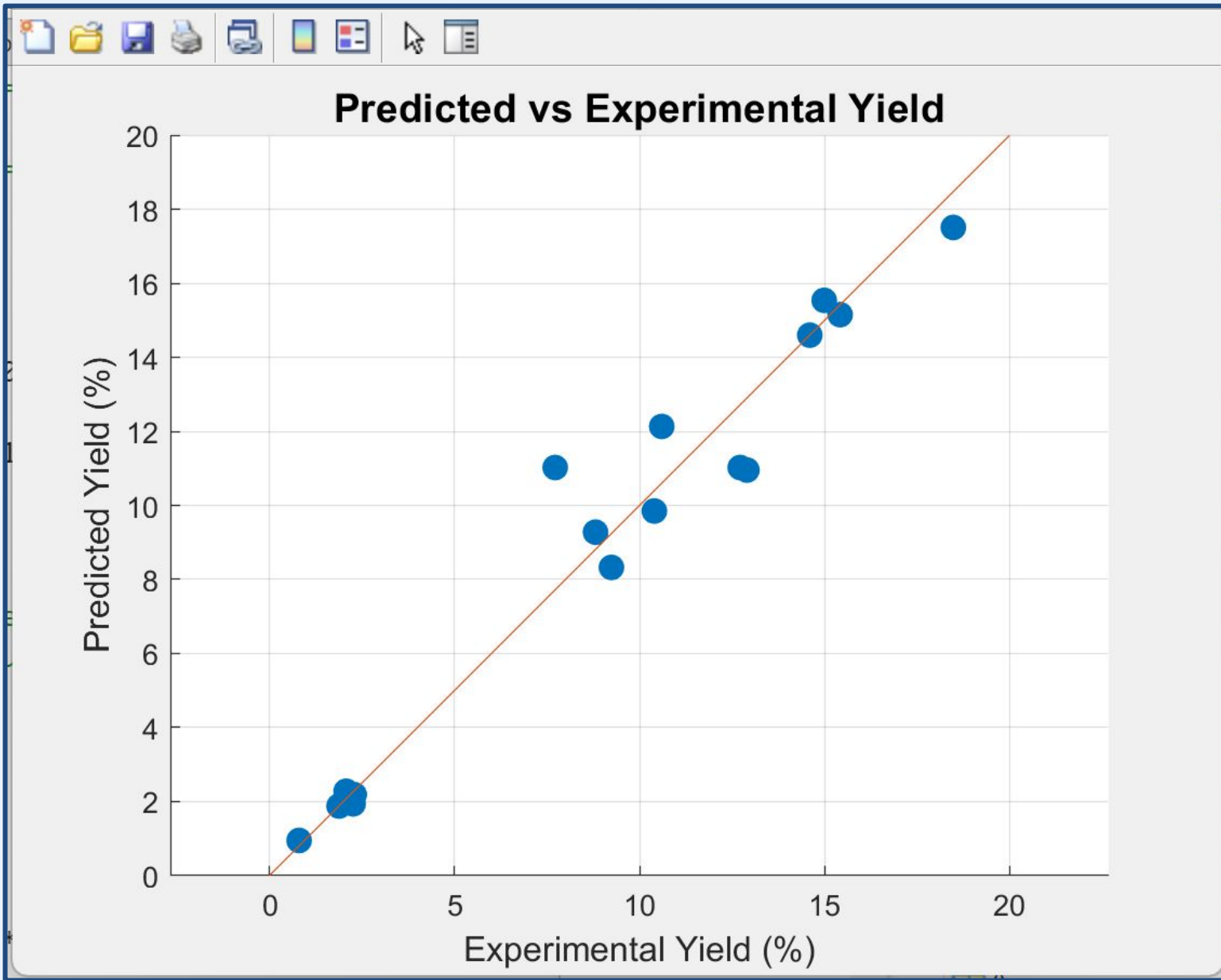
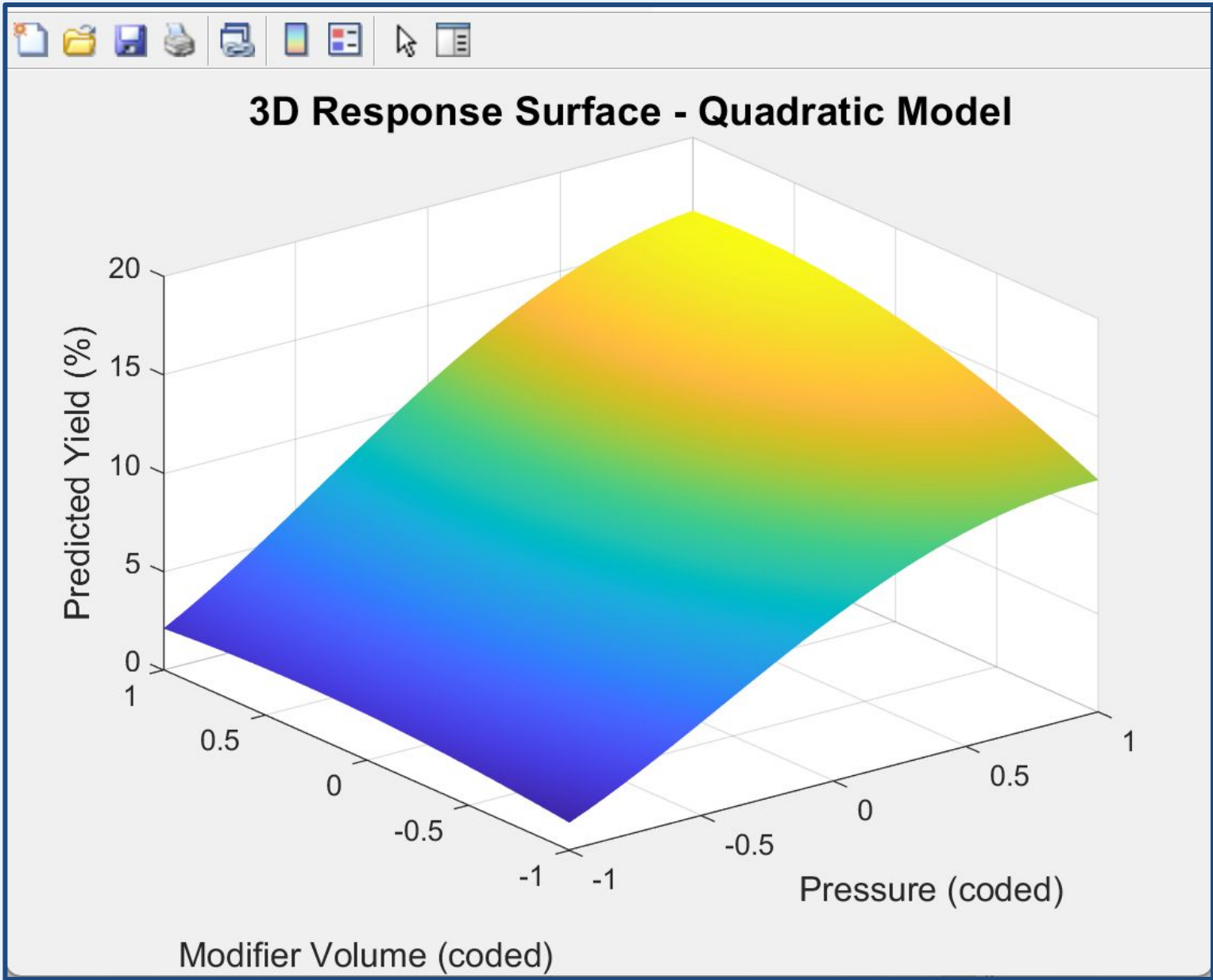


Applications

- Used in aromatherapy for relaxation and stress reduction.
- Acts as a natural antimicrobial agent in household cleaners.
- Widely used in skincare products for its refreshing fragrance and antibacterial properties.
- Added to food and beverages as a natural flavoring agent.
- Applied in herbal medicine for pain relief and inflammation reduction.
- Used in perfumery for creating citrus based fragrances.



Observations and Results



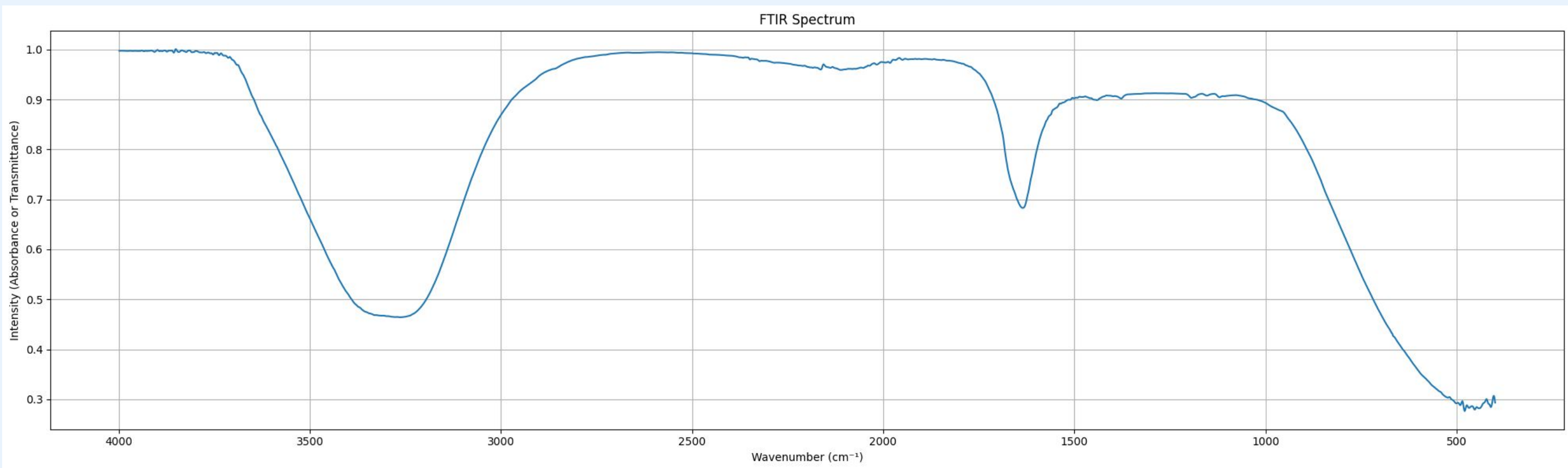
Fitted model equation:

$$\text{Sqrt}(Y) = 3.310 + 1.207A + 0.218B + 0.173C + 0.085AB + 0.094AC + -0.121BC + -0.575A^2 + -0.207B^2 + 0.001C^2$$

where A, B, C are Pressure(bar), Modifier Volume(μL) and Dynamic extraction time(min).

R-squared: 0.9535

FTIR Results



Citral (C=O at $\sim 1700\text{ cm}^{-1}$) Benefits: Antimicrobial, antifungal, antioxidant. Confirmed by: Strong carbonyl peak near 1700 cm^{-1} .

Terpenes (C H at $2850\text{ to }3000\text{ cm}^{-1}$, C=C at $\sim 1650\text{ cm}^{-1}$) Benefits: Antibacterial, antiviral, decongestant. Confirmed by: Clear alkene C H and C=C absorption peaks.

Terpene Alcohols (O H at $3300\text{ to }3500\text{ cm}^{-1}$) Benefits: Skin healing, antiseptic, soothing effects. Confirmed by: Broad O H stretching band around 3300 cm^{-1} .

Esters or C O compounds ($1000\text{ to }1200\text{ cm}^{-1}$) Benefits: Mild anti inflammatory action and fragrance. Confirmed by: Strong C O stretching absorption in this region.

Concluding Remarks

- Lemongrass essential oil was successfully extracted using steam distillation with clear phase separation and measurable yield.
- FTIR analysis confirmed the presence of key functional groups including aldehydes, alkenes, alcohols, and oxygenated compounds.
- These groups correspond to major bioactive components such as citral and terpenes, validating the quality of the extracted oil.
- The confirmed components exhibit strong antimicrobial, antioxidant, and therapeutic benefits, supporting the natural value of lemongrass oil.
- Overall, the experiment demonstrated efficient extraction, reliable characterization, and consistency with known chemical and biological properties of lemongrass essential oil.

Contributions

- Aftab - Process Design, Experimental Setup and FTIR Analysis
- Yashvi - Quadratic Model Fitting, Experimental Execution & Analysis, FTIR Characterization
- Prakhar- Experimental Set-up, Preparation, Execution and primary Data Analysis
- Pradeep- Arrangement of feed sample, experimental execution and analysis
- Sai Tejeshwar- Experimental setup, Conducting multiple trials and Analysis of the collected data

References

- [1] F. Teshale, K. Narendiran, S. M. Beyan, and N. R. Srinivasan, "Extraction of essential oil from rosemary leaves: optimization by response surface methodology and mathematical modeling," *Applied Food Research*, vol. 2, p. 100133, 2022.
- [2] K. M. Ara, M. Jowkarderis, and F. Raofie, "Optimization of supercritical fluid extraction of essential oils and fatty acids from flaxseed (*Descurainia sophia* L.) seed using response surface methodology and central composite design," *Journal of Food Science and Technology*, vol. 52, no. 7, pp. 4450 to 4458, 2015.